



HABIB UNIVERSITY

DHANANI SCHOOL OF SCIENCE AND ENGINEERING

EE251/CE252	Spring'26	HW 4 SOLUTION	Marks 50 Weight 2.5%	Due: As announced
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Important Notes

Preceding the solutions, you must include the following information:

- Your name
- Student ID
- Course ID
- Homework number
- Date and time submitted

Solution should be elaborative, sketch neat circuits wherever required, show clearly all steps of calculation & mention reference (for example formula, law, rule) to strengthen your understanding to solve the electric circuit problems.

Also, please comply with Academic Integrity Policies of Habib University related to plagiarism while submitting this assignment for grading.

Late Submission Policy

As announced in course syllabus.

CLO-4	Analyze CT signals and systems in the transform domain using Laplace, Fourier transforms and Fourier Series.	PLO-2 (Problem Analysis)	Cognitive Level-4 (Analyzing)
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Problem 1 [10 pts]

Find the inverse Laplace transforms of the following functions:

- (a) $\frac{(2s+5)e^{-2s}}{s^2+5s+6}$
- (b) $\frac{se^{-3s}+2}{s^2+2s+2}$

Solution:

(a)

$$X(s) = \frac{(2s+5)e^{-2s}}{s^2+5s+6} = \hat{X}(s)e^{-2s}$$

It is clear that $x(t) = \hat{x}(t-2)$.

$$\hat{X}(s) = \frac{2s+5}{s^2+5s+6} = \frac{2s+5}{(s+2)(s+3)} = \frac{1}{s+2} + \frac{1}{s+3}$$

$$\hat{x}(t) = (e^{-2t} + e^{-3t})u(t)$$

$$x(t) = \hat{x}(t-2) = [e^{-2(t-2)} + e^{-3(t-2)}]u(t-2)$$

(b)

$$X(s) = \frac{s}{s^2+2s+2}e^{-3s} + \frac{2}{s^2+2s+2} = X_1(s)e^{-3s} + X_2(s)$$

where

$$X_1(s) = \frac{s}{s^2+2s+2} \quad \left\{ \begin{array}{l} A=1, B=0, a=1, c=2, b=1 \\ r=\sqrt{2}, \theta=\tan^{-1}(1)=\pi/4 \end{array} \right.$$

$$x_1(t) = \sqrt{2}e^{-t} \cos(t + \frac{\pi}{4})$$

$$X_2(s) = \frac{2}{s^2+2s+2} \quad \text{and} \quad x_2(t) = 2e^{-t} \sin t$$

Also

$$\begin{aligned} x(t) &= x_1(t-3) + x_2(t) \\ &= \sqrt{2}e^{-(t-3)} \cos(t-3 + \frac{\pi}{4})u(t-3) + 2e^{-t} \sin t u(t) \end{aligned}$$

Problem 2 [5 pts]

Consider a causal LTIC system described by the differential equation

$$2\dot{y}(t) + 6y(t) = \dot{x}(t) - 4x(t)$$

- (a) Using transform-domain techniques, determine the ZIR $y_{\text{zir}}(t)$ if $y(0^-) = -3$.
- (b) Using transform-domain techniques, determine the ZSR $y_{\text{zsr}}(t)$ to the input $x(t) = e\delta(t - \pi)$.

Solution:

- (a) Setting $x(t) = 0$ and taking the unilateral Laplace transform of $2\dot{y}(t) + 6y(t) = \dot{x}(t) - 4x(t)$ yield

$$2[sY_{\text{zir}}(s) - y(0^-)] + 6Y_{\text{zir}}(s) = 0.$$

Thus,

$$Y_{\text{zir}}(s) = \frac{y(0^-)}{s+3} = \frac{-3}{s+3}.$$

Inverting,

$$y_{\text{zir}}(t) = -3e^{-3t}u(t).$$

- (b) The input is $x(t) = e\delta(t - \pi)$. Using entry 1 of Table 4.1 and the time-shifting property,

$$X(s) = ee^{-s\pi}.$$

By inspection of $2\dot{y}(t) + 6y(t) = \dot{x}(t) - 4x(t)$, the system transfer function is

$$H(s) = \frac{\frac{1}{2}s - 2}{s + 3}.$$

The transform of the zero-state response is

$$Y_{\text{zsr}}(s) = X(s)H(s) = ee^{-s\pi} \frac{\frac{1}{2}s - 2}{s + 3} = ee^{-s\pi} \left[\frac{1}{2} + \frac{-\frac{7}{2}}{s + 3} \right].$$

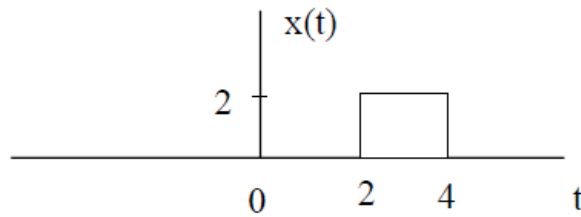
Inverting,

$$y_{\text{zsr}}(t) = e \left[-\frac{7}{2}e^{-3(t-\pi)}u(t-\pi) + \frac{1}{2}\delta(t-\pi) \right].$$

Problem 3 [10 pts]

Find the Fourier transform of the following signals. Sketch $|X(\omega)|$ in each case.

a)



b) $x(t) = 2e^{-2t}u(t)$

c) $x(t) = 5e^{-5t}u(t)$

d) $x(t) = e^{-2t}\cos(4t)u(t)$

SOLUTION

Let $p_a(t)$ be a rectangular pulse of width a centered at the origin.

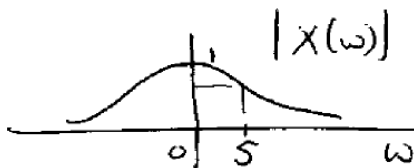
a) $x(t) = 2p_2(t-3) \longleftrightarrow 4 \operatorname{sinc}\left(\frac{\omega}{2\pi}\right) e^{-3j\omega} = X(\omega)$



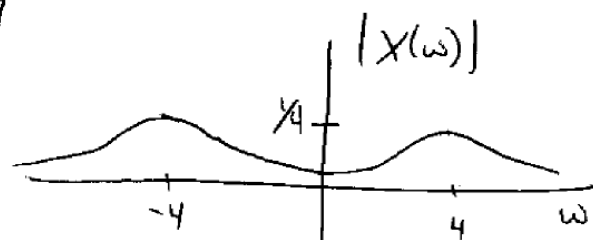
b) $X(\omega) = \frac{2}{2+j\omega}$



c) $X(\omega) = \frac{5}{5+j\omega}$



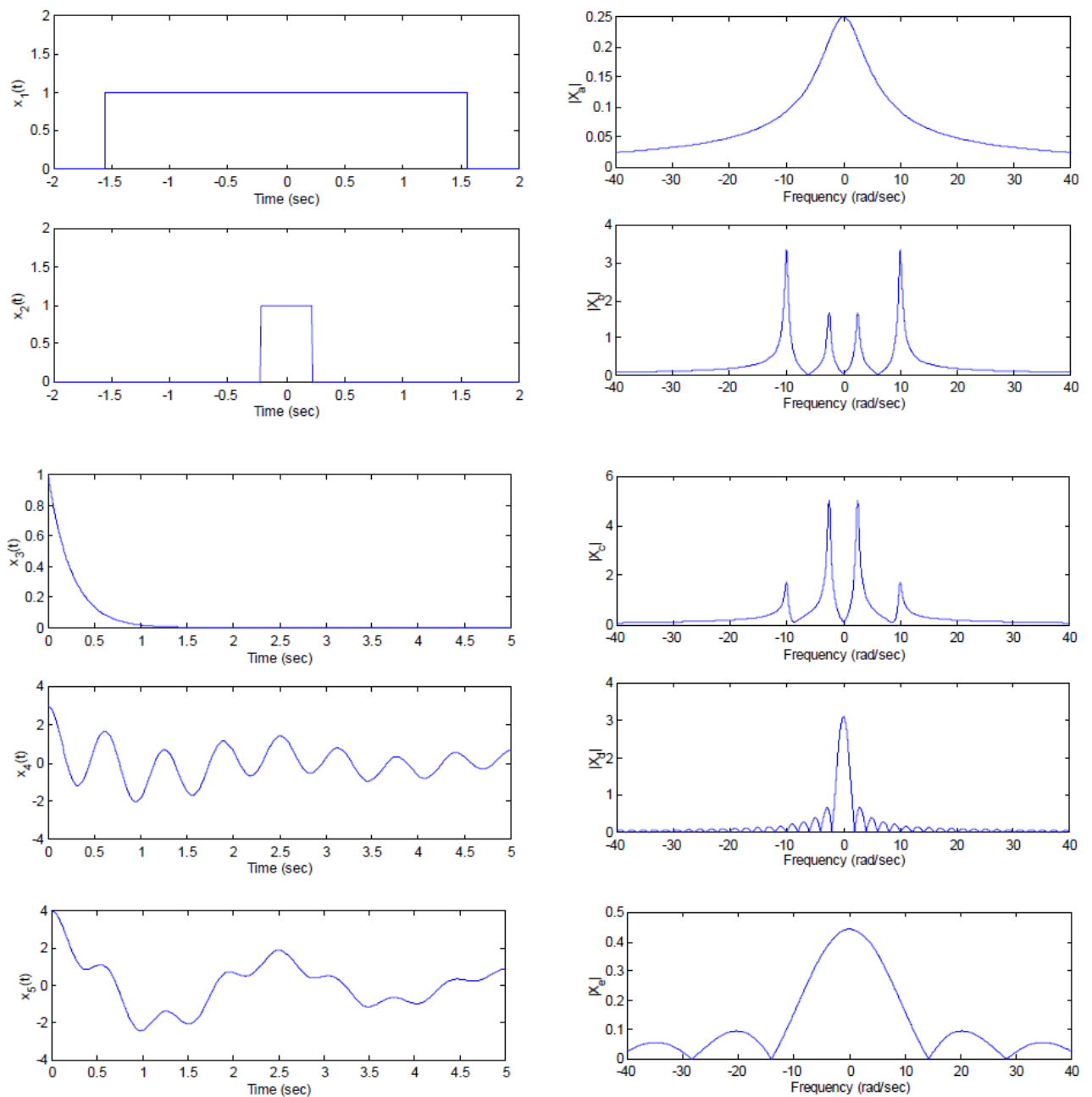
d) $X(\omega) = \frac{1}{2} \left[\frac{1}{j(\omega-4)+2} + \frac{1}{j(\omega+4)+2} \right]$



Problem 4 [5 pts]

Match the time responses with the corresponding frequency responses.

(justify your choices)



SOLUTION

1. d 2. e 3. a 4. b 5. c

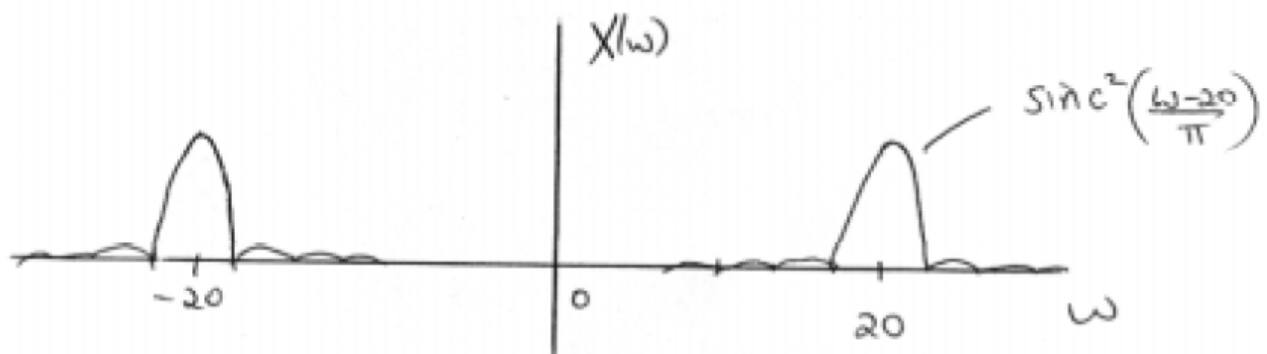
Reasoning:

- FT of Rectangle is Sinc function. Spread in one domain causes narrowing in the other. These lead to the pairing 1-d and 2-e.

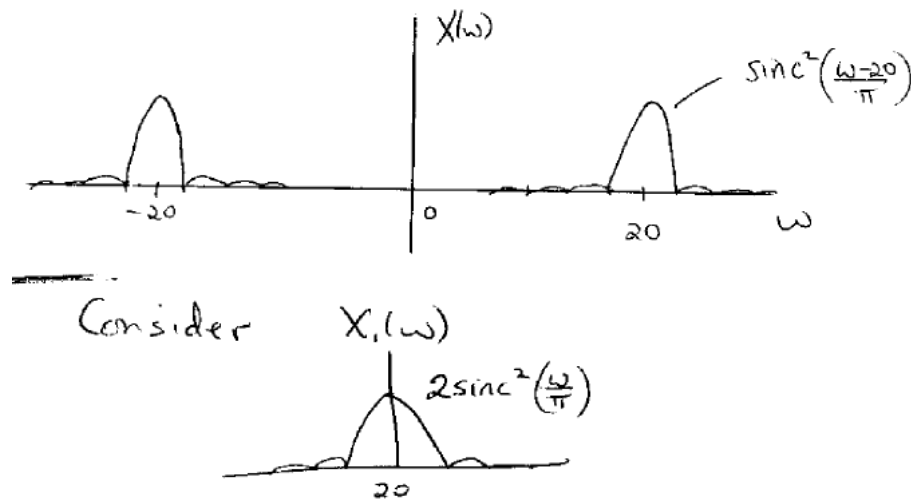
- DC has a peak at zero, damped DC should lead to spreading of the peak at zero. This gives the pairing 3-a.
- 4 and 5 seem to be made of a small set of sinusoids. In 4 overall high frequency seems dominant, while in 5 low frequency seems dominant, leading to the pairing 4-b and 5-c.

Problem 5 [10 pts]

Compute the inverse Fourier transform of the following signal.



SOLUTION



Now the given Frequency signal $X(\omega)$ is a shifted version of $X_1(\omega)$ (shifted by ± 20), i.e., we have

$$X(\omega) = \frac{1}{2} X_1(\omega - 20) + \frac{1}{2} X_1(\omega + 20)$$

Applying the Frequency Shifting property to the above, leads to

$$x(t) = \frac{1}{2} x_1(t) e^{j20t} + \frac{1}{2} x_1(t) e^{-j20t}$$

Using Euler's identity, this reduces to

$$x(t) = x_1(t) \cos(20t)$$

Where

$$x_1(t) = \Delta\left(\frac{t}{4}\right)$$

Problem 6 [10 pts]

An **Exponential Fourier Series** is defined as

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$$

Whereas, a **Trigonometric Fourier Series** is defined as

$$x(t) = \frac{a_0}{2} + \sum_{k=1}^{\infty} (a_k \cos k\omega_0 t + b_k \sin k\omega_0 t)$$

Find the relationship between the coefficients (c_k, a_0, a_k, b_k) of the two series.

SOLUTION

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t} = c_0 + \sum_{k=1}^{\infty} (c_k e^{jk\omega_0 t} + c_{-k} e^{-jk\omega_0 t})$$

and using Euler's formulas

$$e^{\pm jk\omega_0 t} = \cos k\omega_0 t \pm j \sin k\omega_0 t$$

we have

$$x(t) = c_0 + \sum_{k=1}^{\infty} [(c_k + c_{-k}) \cos k\omega_0 t + j(c_k - c_{-k}) \sin k\omega_0 t]$$

Setting

$$c_0 = \frac{a_0}{2} \quad c_k + c_{-k} = a_k \quad j(c_k - c_{-k}) = b_k$$

The Exponential Fourier Series above becomes

